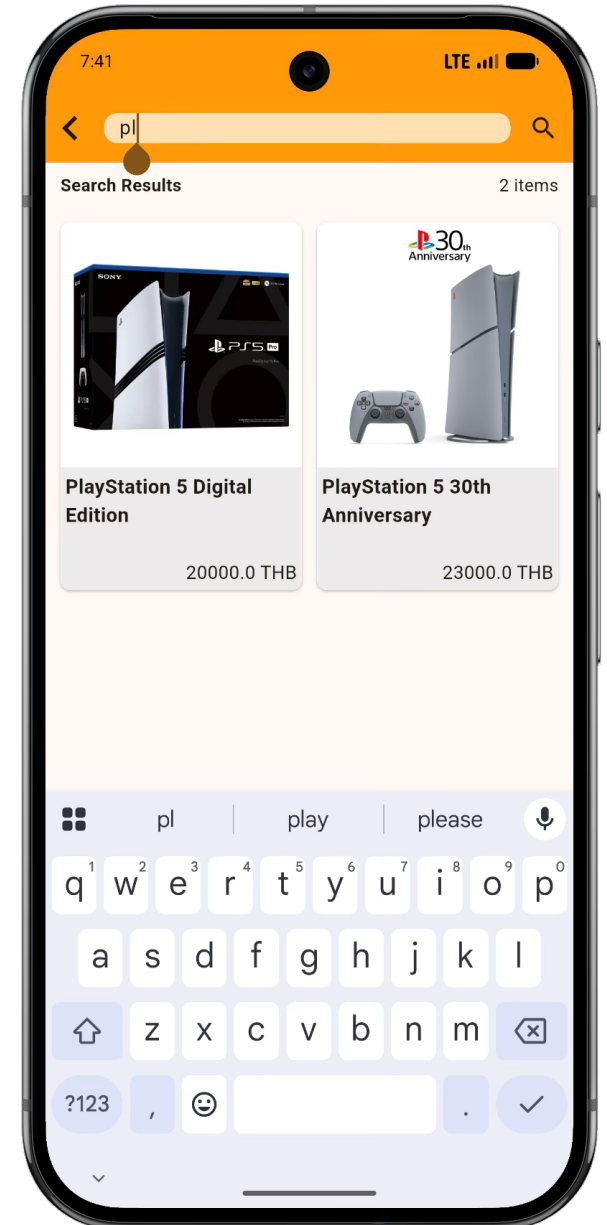


Homework 13

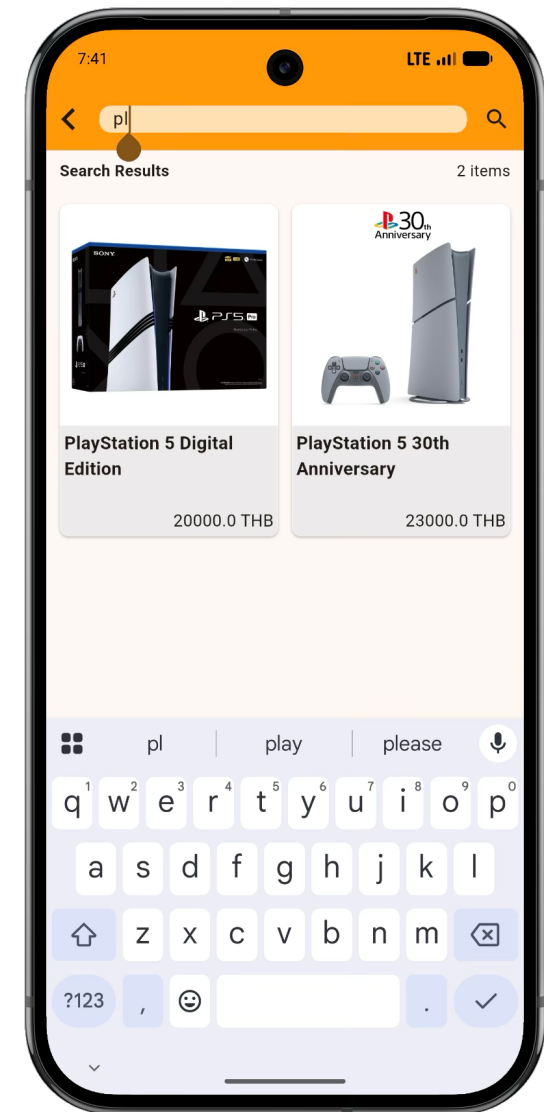
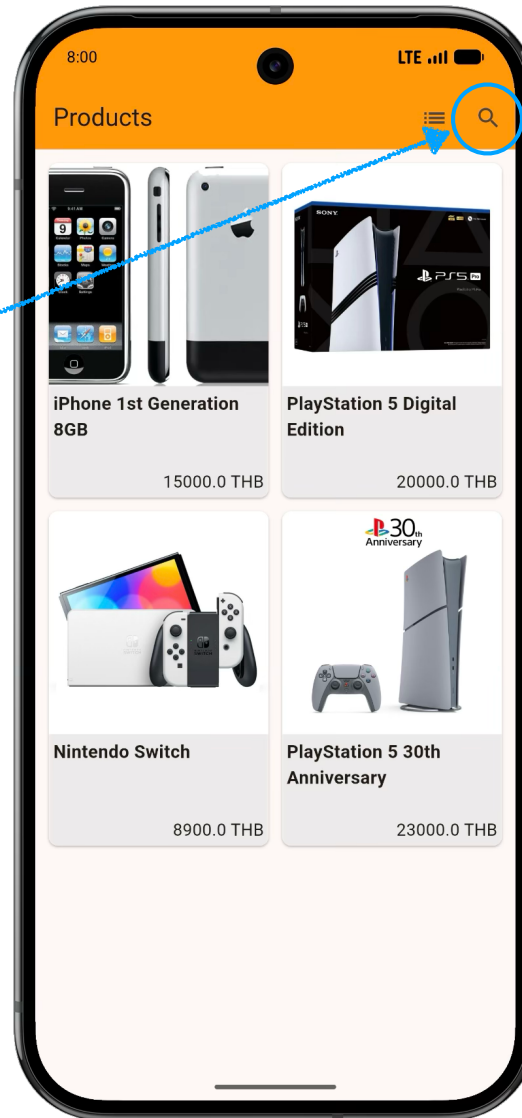
- Complete your class activity.
- Implement search product function.



Homework 13

Search Products with Keywords

Search Product Items
With Keywords

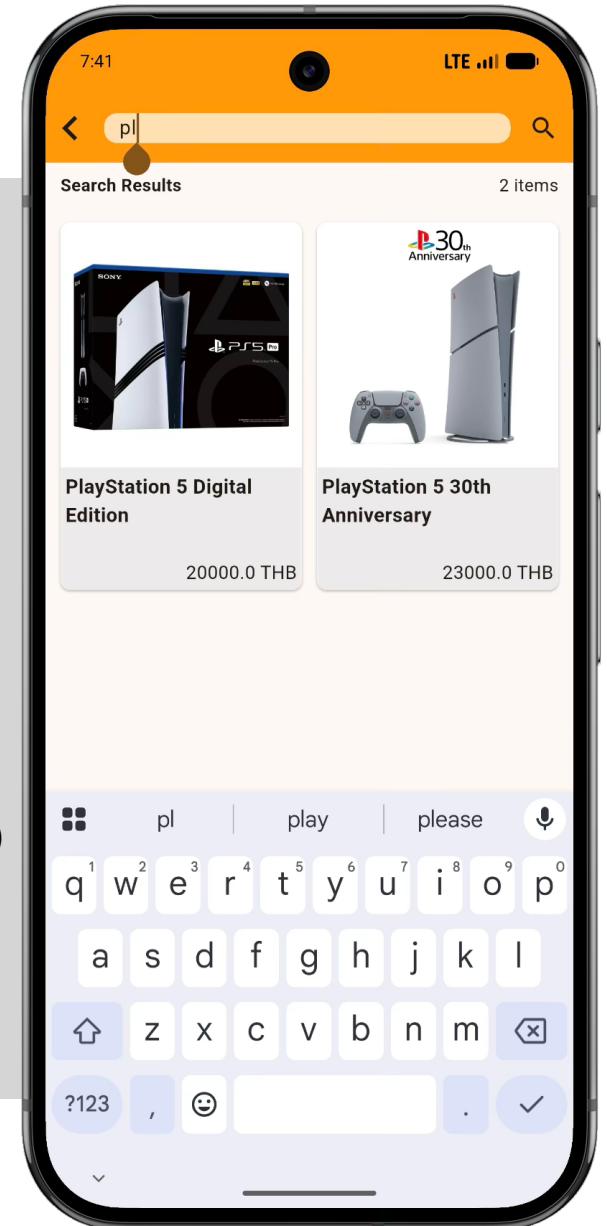


Homework 13

Add searchProducts in ProductManager

- Add these lines of code in **product_manager.dart** for searching products

```
class ProductManager {  
  ...  
  
  Future<List<ProductModel>> searchProducts(String keyword) async {  
    if (keyword.isEmpty) {  
      return [];  
    }  
  
    final items = await getProducts();  
  
    return items  
      .where((item) => item.name.toLowerCase().contains(keyword.toLowerCase()))  
      .toList();  
  }  
}
```

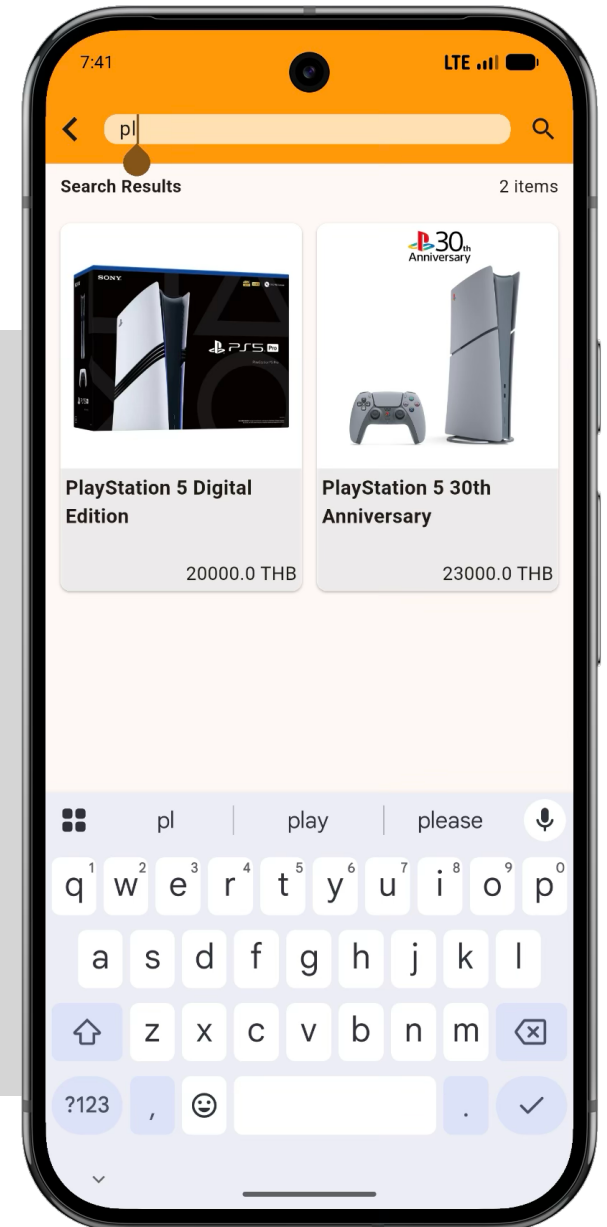


Homework 13

Product Search Screen

- Create a stateful widget **'product_search_screen.dart'** in products directory. This screen allows user find desired product items with keywords.

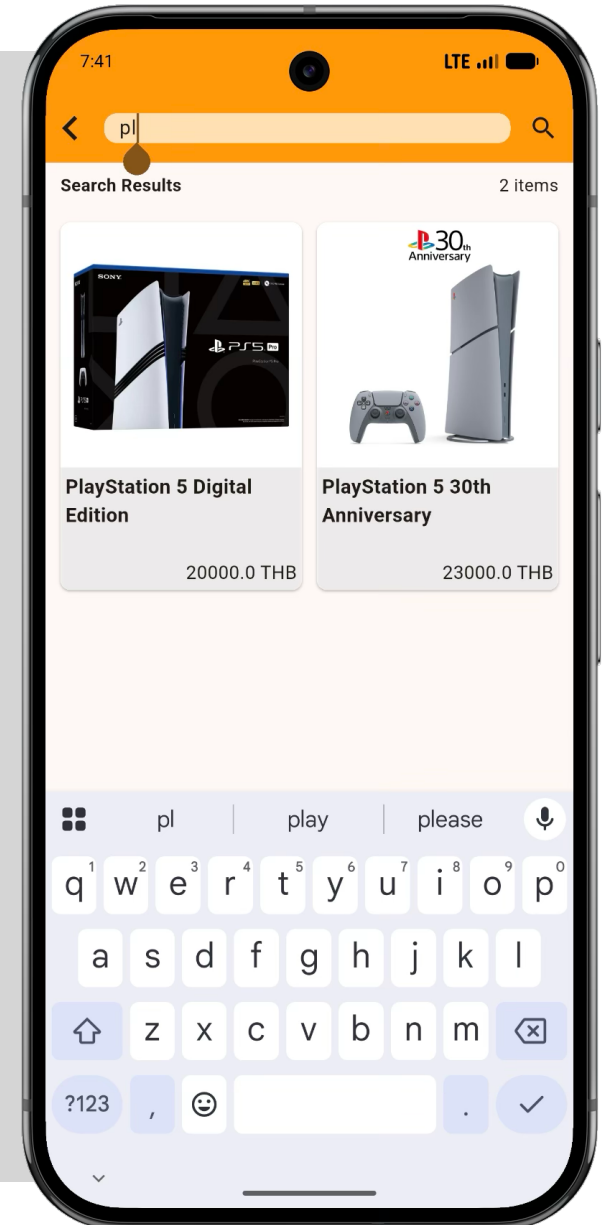
```
class ProductSearchScreen extends StatefulWidget {  
  ...  
}  
  
class _ProductSearchScreenState extends State<SearchScreen> {  
  final _formKey = GlobalKey<FormState>();  
  final _productManager = ProductManager();  
  Future<List<ProductModel>>? _result;  
  late TextEditingController _keywordController;  
  ...  
}
```



Homework 13

Product Search Screen

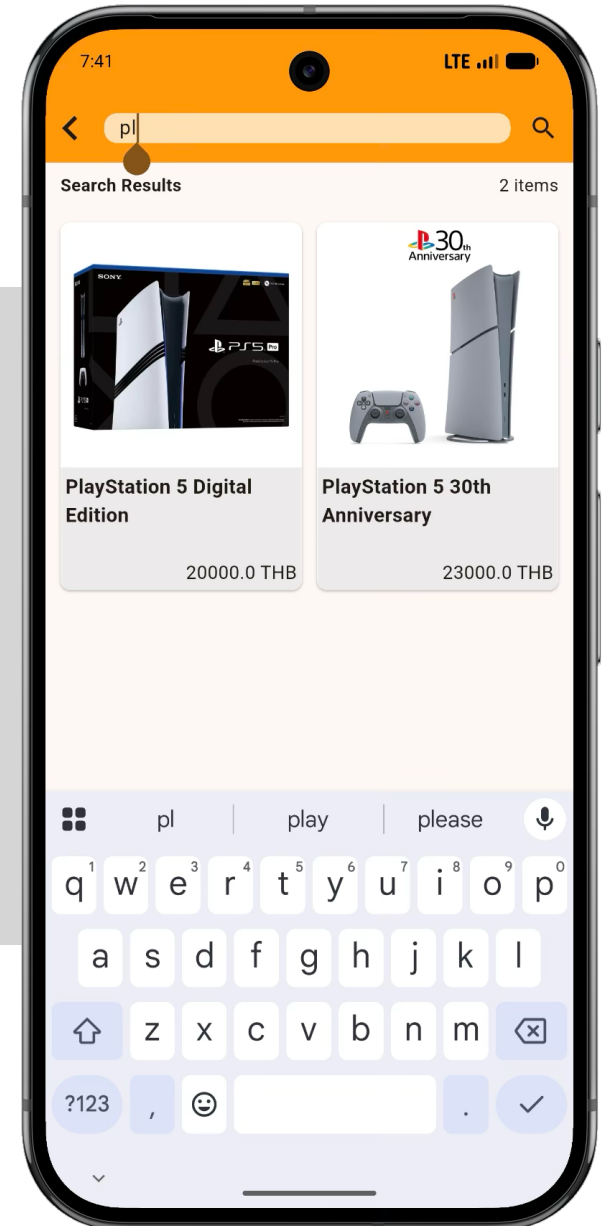
```
class _ProductSearchScreenState extends State<ProductSearchScreen> {  
  ...  
  
  Future<List<ProductModel>> searchProduct(String keyword) async {  
    return await _productManager.searchProducts(keyword);  
  }  
  
  @override  
  void initState() {  
    super.initState();  
    _keywordController = TextEditingController();  
  }  
  
  @override  
  void dispose() {  
    _keywordController.dispose();  
    super.dispose();  
  }  
  ...  
}
```



Homework 13

Product Search Screen

```
class _ProductSearchScreenState extends State<ProductSearchScreen> {  
  ...  
  @override  
  Widget build(BuildContext context) {  
    return Scaffold(  
      appBar: ...  
      body: ...  
    );  
  }  
  ...  
}
```



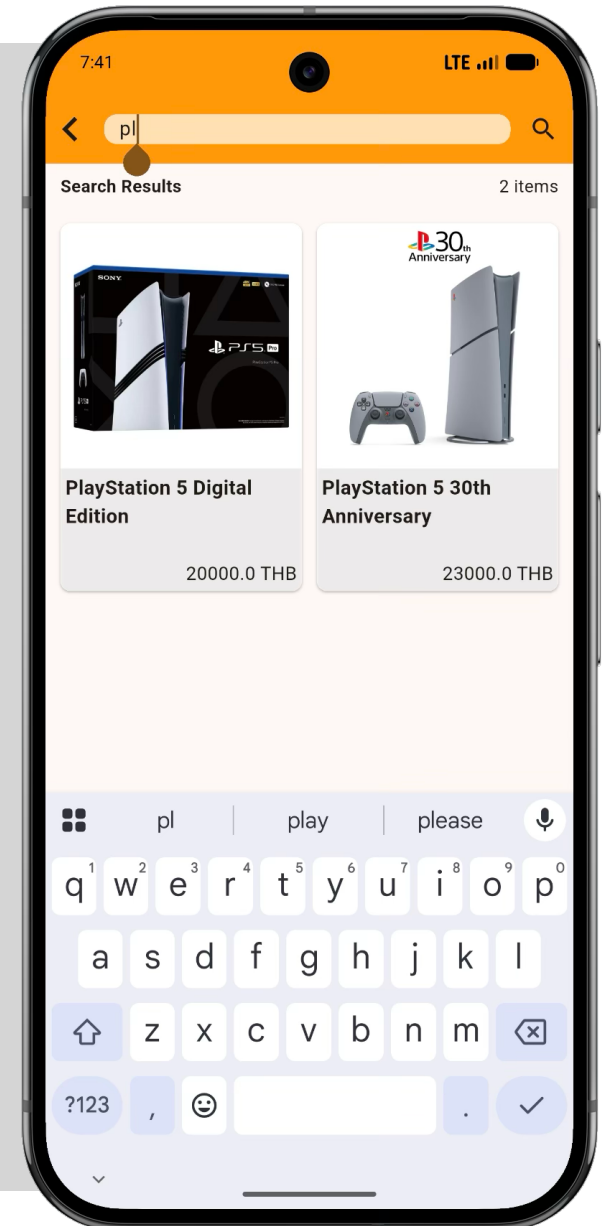
Homework 13

Product Search Screen

```

appBar: AppBar(
  title: Row(
    mainAxisAlignment: MainAxisAlignment.spaceBetween,
    children: [
      Flexible( flex: 1,
        child: GestureDetector(
          onTap: () => Navigator.of(context).pop(),
          child: Icon(Icons.keyboard_arrow_left, size: 40),
        ),
      Expanded( flex: 10,
        child: Form(
          key: _formKey,
          child: TextFormField(
            controller: _keywordController,
            autofocus: true,
            keyboardType: TextInputType.text,
            autocorrect: false,
            onFieldSubmitted: (keyword) =>
              onFieldSubmitted: (keyword) {
                setState(() {
                  _result = searchProduct(keyword);
                });
              },
            decoration: InputDecoration.collapsed(
              hintText: 'Search',
              fillColor: Colors.white70,
              filled: true,
              border: OutlineInputBorder(
                borderRadius: BorderRadius.circular(10.0),
                borderSide: BorderSide.none,
                gapPadding: 12,
              ),
            ),
          ),
        ),
      Flexible( flex: 2,
        child: GestureDetector(
          onTap: () {
            if (_formKey.currentState!.validate()) {
              setState(() {
                _result = searchProduct(_keywordController.text);
              });
            }
          },
          child: Padding(
            padding: const EdgeInsets.only(right: 12.0),
            child: Icon(Icons.search),
          ),
        ),
      ),
    ],
  ),
  automaticallyImplyLeading: false,
  leadingWidth: 0,
  titleSpacing: 0,
  body: ...

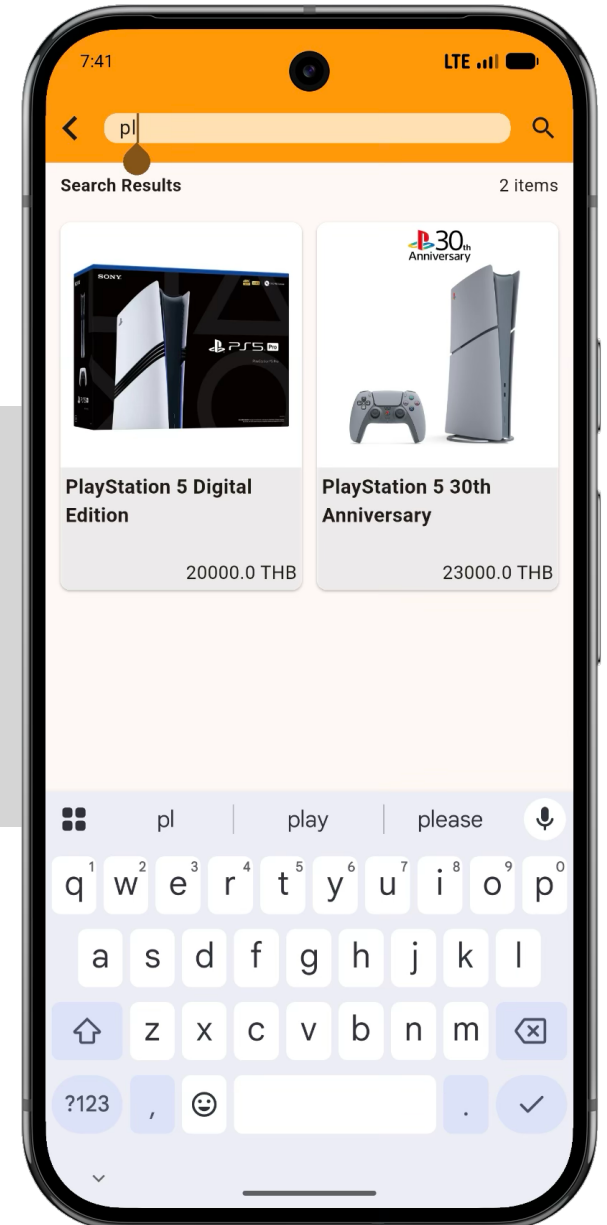
```



Homework 13

Product Search Screen

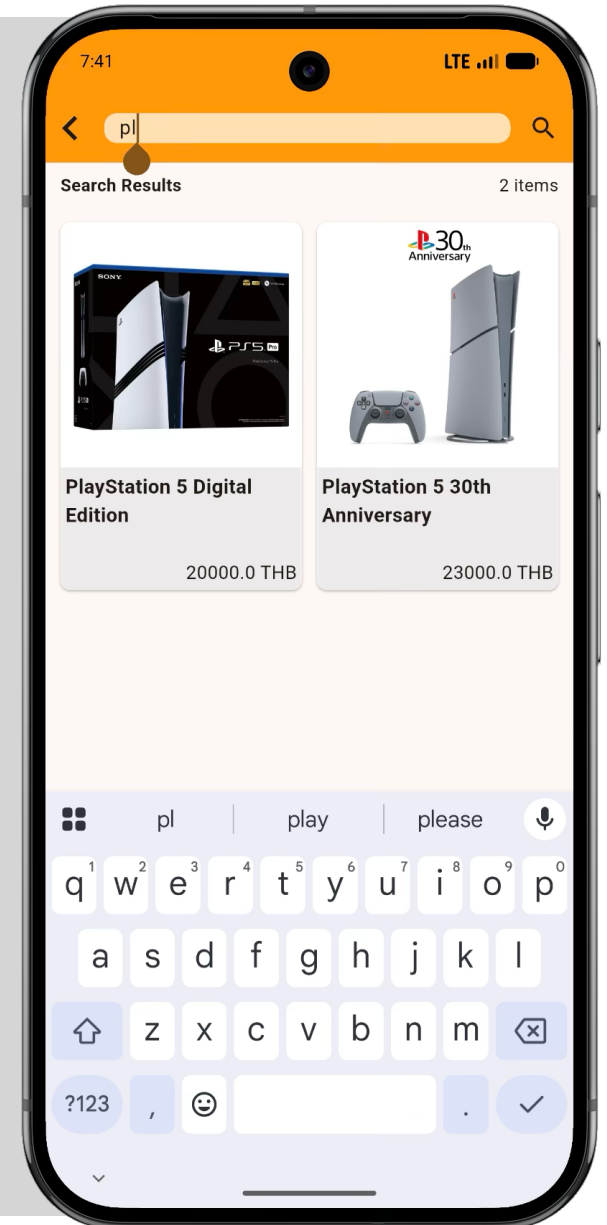
```
appBar: ...  
body: FutureBuilder<List<ProductModel>>(  
  future: _result,  
  builder: (context, snapshot) {  
    ...  
  }  
)
```



Homework 13

Product Search Screen

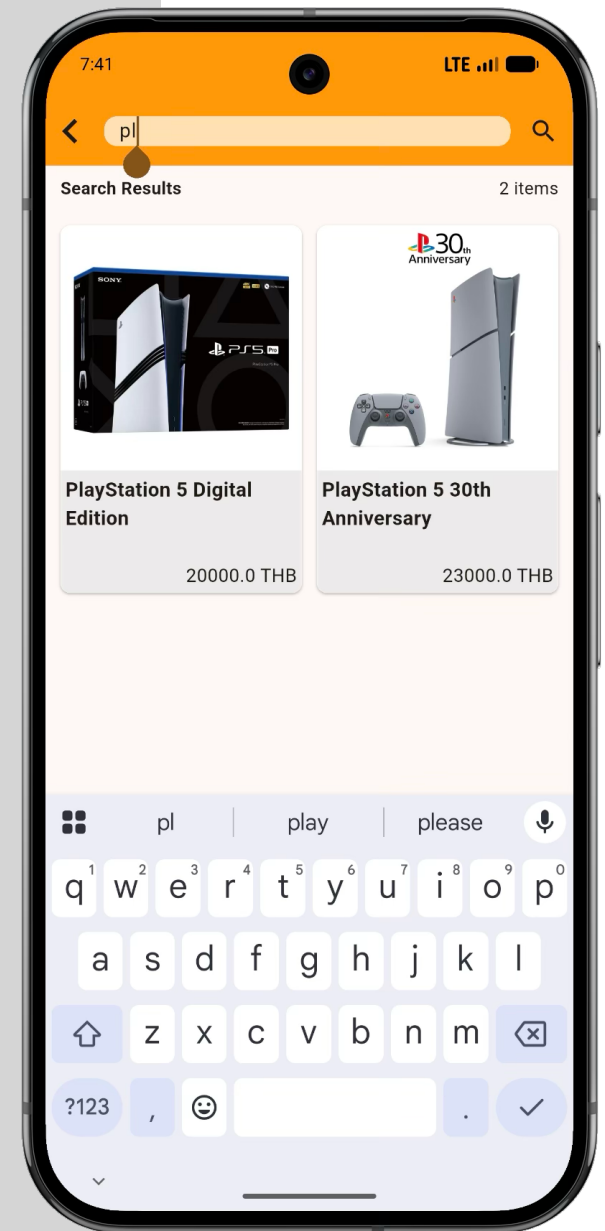
```
appBar: ...
body: FutureBuilder<List<ProductModel>>(
  future: _result,
  builder: (context, snapshot) {
    // Loading State
    if (snapshot.connectionState == ConnectionState.waiting) {
      return Column(
        mainAxisAlignment: MainAxisAlignment.center,
        crossAxisAlignment: CrossAxisAlignment.stretch,
        children: [
          Center(
            child: CircularProgressIndicator(color: Colors.grey.shade500),
          ),
          SizedBox(height: 12),
          Text("Searching...",
            textAlign: TextAlign.center,
            style: TextStyle(fontSize: 20, color: Colors.grey.shade500),
          ),
        ],
      );
    }
    ...
  }
),
```



Homework 13

Product Search Screen

```
appBar: ...
body: FutureBuilder<List<ProductModel>>(
  future: _result,
  builder: (context, snapshot) {
    ....
    // Handling Error State ...
    if (snapshot.hasError) {
      return Column(
        mainAxisAlignment: MainAxisAlignment.center,
        crossAxisAlignment: CrossAxisAlignment.stretch,
        children: [
          Icon(
            Icons.error_outline_rounded,
            size: 100,
            color: Colors.grey.shade500,
          ),
          SizedBox(height: 12),
          Text(
            "Could not search products.",
            textAlign: TextAlign.center,
            style: TextStyle(fontSize: 20, color: Colors.grey.shade500),
          ),
        ],
      );
    }
    ...
  },
),
```

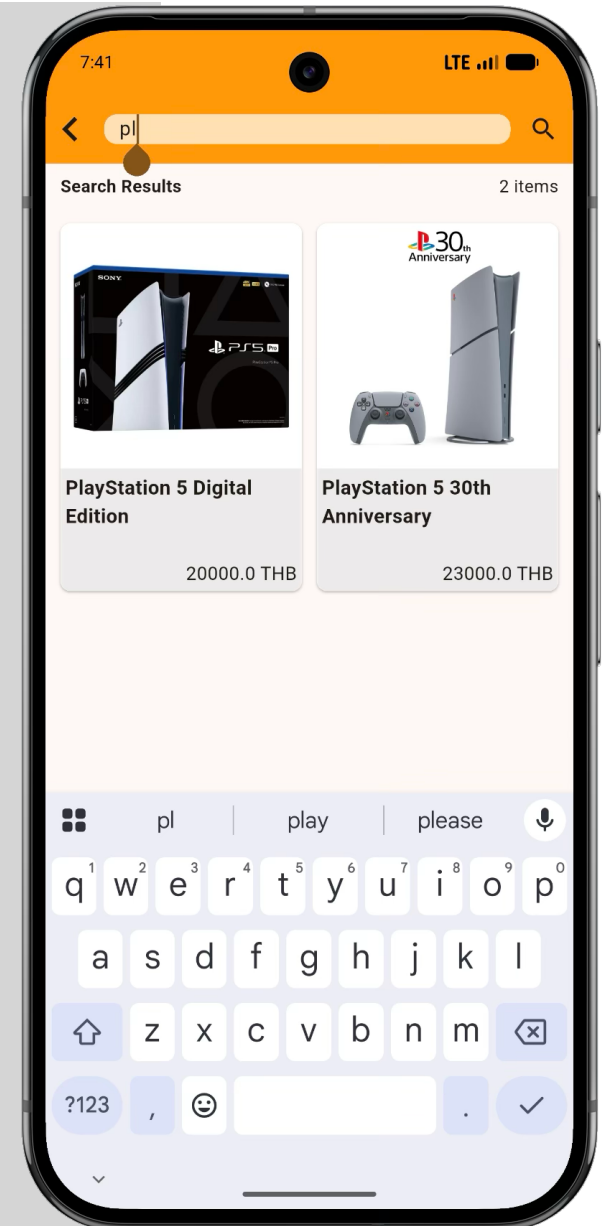


Homework 13

Product Search Screen

```
appBar: ...
body: FutureBuilder<List<ProductModel>>({
  future: _result,
  builder: (context, snapshot) {
    ...
    // Extract Search Result ...
    final searchResult = snapshot.data ?? [];

    if (searchResult.isEmpty) {
      return Column(
        mainAxisAlignment: MainAxisAlignment.center,
        crossAxisAlignment: CrossAxisAlignment.stretch,
        children: [
          Icon(
            Icons.search_off_rounded,
            size: 100,
            color: Colors.grey.shade500,
          ),
          Text(
            "No Results",
            textAlign: TextAlign.center,
            style: TextStyle(fontSize: 20, color: Colors.grey.shade500),
          ),
        ],
      );
    }
    ...
  },
),
```



Homework 13

Product Search Screen

```
appBar: ...
body: FutureBuilder<List<ProductModel>>{
  future: _result,
  builder: (context, snapshot) {
    ...
    // Display Result. ...
    return Column(
      crossAxisAlignment: CrossAxisAlignment.stretch,
      children: [
        Padding(
          padding: const EdgeInsets.symmetric(
            horizontal: 12.0,          vertical: 8.0,
          ),
          child: Row(
            mainAxisAlignment: MainAxisAlignment.spaceBetween,
            children: [
              Text(
                "Search Results",
                style: TextStyle(fontWeight: FontWeight.bold),
              ),
              Text("${searchResult.length} items"),
            ],
          ),
          Expanded(
            child: productInGrid(searchResult)
          ),
        ],
      );
    }
  ),
),
```

